

SANDEEP NIMMA

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SUMMARY

- Software Development Engineer with 5 years of expertise in designing, developing, and maintaining scalable, cloud-native applications using Java, Spring Boot, React.js, and microservices architecture.
- Skilled in building robust backend systems, implementing RESTful APIs, and integrating real-time data processing and inventory management solutions for high-performance applications.
- Proficient in designing and executing automated test frameworks using Selenium WebDriver, TestNG, JUnit, and REST Assured, ensuring high-quality software releases and reliable application performance.
- Hands-on experience in CI/CD pipeline implementation using Jenkins, Docker, and Kubernetes, enabling efficient deployment workflows and seamless integration between development, testing, and production environments.
- Strong expertise in working with relational and NoSQL databases such as MongoDB, DynamoDB, Oracle, and SQL Server, including query optimization, indexing, and schema design for large-scale applications.
- Adept at monitoring and logging using ELK Stack, CloudWatch, and Splunk, proactively identifying performance bottlenecks and maintaining application stability.
- Collaborative team player with experience working in Agile environments, coordinating with cross-functional teams using Jira and Confluence to deliver iterative product enhancements.
- Well-versed in cloud platforms including AWS, Azure, and GCP, leveraging cloud-native services to build scalable, secure, and maintainable enterprise solutions.

SKILLS

Programming & Scripting: Java, C#, JavaScript (ES6+), TypeScript, SQL

Web & Front-End: HTML5, CSS3, React.js, Angular, Vue.js, Bootstrap, Tailwind CSS, RESTful APIs, GraphQL

Back-End & Frameworks: Spring Boot, Node.js, .NET Core, Microservices, Serverless, Express.js

Testing & Automation: Selenium WebDriver, Cypress, JUnit, NUnit, TestNG, RestAssured, Postman, JMeter, LoadRunner, Playwright

CI/CD & DevOps: Jenkins, GitLab CI/CD, GitHub Actions, CircleCI, Bamboo, Docker, Kubernetes, Ansible, Terraform, Maven, Gradle

Version Control & Collaboration: Git, GitHub, GitLab, Bitbucket, Jira, Confluence, Agile & Scrum

Cloud & Platforms: AWS (EC2, S3, Lambda, CloudFormation), Azure DevOps, GCP, Serverless Framework, CloudWatch

Databases & Data Management: MySQL, PostgreSQL, MongoDB, Redis, Oracle, SQL Server, DynamoDB, BigQuery, Snowflake

Test Management & Reporting: Zephyr, TestRail, qTest, Allure, ExtentReports, SonarQube, Jenkins Reporting

Automation Frameworks & Methodologies: BDD/TDD frameworks, Page Object Model (POM), Data-Driven & Keyword-Driven Testing, Selenium Grid, Robot Framework

Monitoring & Performance Tools: ELK Stack, Grafana, Prometheus, New Relic, Splunk, AppDynamics, JMeter, LoadRunner

Other Technical Expertise: REST API Automation, API Mocking & Virtualization, Cloud-Native Testing, Dockerized Test Environments, Performance & Security Testing, Test Automation Strategy Design

WORK EXPERIENCE

Nationwide

Software Development Engineer

Ohio, USA

May 2024 – Present

- Collaborated with business analysts and product owners to gather complex insurance rating requirements and translated them into technical specifications, ensuring compliance and accurate processing for over 100,000 daily policies.
- Refactored monolithic rating logic into modular Java 11 and Spring Boot microservices, enabling independent deployment of new insurance products and reducing release timelines from three weeks to ten days while maintaining system stability.
- Designed and implemented RESTful APIs for real-time integration between rating engines and policy management systems, improving transaction throughput and enabling accurate pricing calculations across multiple insurance lines simultaneously.
- Leveraged Apache Kafka for asynchronous messaging between microservices, supporting high-volume transaction processing and reducing API response latency from 1.2 seconds to 0.9 seconds per request, improving overall system efficiency.
- Implemented Redis caching for frequently accessed policy rating data, minimizing database queries and enhancing real-time calculation performance, leading to faster response times for policy evaluation and quote generation.

- Optimized Oracle SQL queries and stored procedures for large-scale rating tables, reducing data retrieval time by 35% and supporting faster computation for complex rate calculations, ensuring system scalability for growing policy volumes.
- Developed and executed comprehensive unit and integration tests using JUnit and Mockito, while automating API testing with Postman to achieve 95% code coverage, significantly reducing post-deployment defects and ensuring system reliability.
- Built and maintained CI/CD pipelines using Jenkins, Maven, and Docker, enabling multiple daily microservice deployments with zero downtime and ensuring seamless integration between development, testing, and production environments.
- Monitored microservices using ELK Stack and Splunk, proactively identifying performance bottlenecks, tracking system health metrics, and reducing downtime incidents by 15%, contributing to consistent SLA adherence.
- Authored detailed API documentation, runbooks, and architecture guides to facilitate cross-team collaboration, knowledge transfer, and maintainable microservices architecture, supporting the team in scaling the system for future enhancements.

Capgemini

India

Software Development Engineer in Test (SDET)

Jun 2021 – Aug 2022

- Analyzed SAP BTP modules and collaborated with developers to define test scenarios and acceptance criteria, ensuring critical workflows, user management features, and analytics functions were fully validated before release
- Designed and developed reusable automated test cases for web applications using Java and Selenium WebDriver, detecting 30% more integration defects during early testing.
- Developed a data-driven automation framework using Java, TestNG, and Maven, supporting multiple datasets and reducing script maintenance by 25%, improving test consistency across releases.
- Integrated automated tests into Jenkins pipelines for nightly execution of 500+ test cases, providing rapid feedback to developers and accelerating release cycles by 20%.
- Executed API testing using REST Assured with Java, validating core SAP BTP service endpoints, verifying response accuracy, performance, and security compliance, contributing to a 40% reduction in post-production defects.
- Developed custom real-time dashboards and reports using ExtentReports, presenting detailed metrics on test coverage, execution status, defect severity trends, and regression results, enabling project stakeholders to make informed decisions.
- Identified, logged, and prioritized defects in Jira, collaborating with developers during daily stand-ups and sprint reviews to ensure timely resolution of critical issues.
- Participated in Java code reviews and automation framework optimization, implementing reusable utilities and best practices, reducing redundant manual testing efforts by 15% per sprint and improving overall team productivity.

HCLTech

India

Associate Software Engineer

Jan 2020 – May 2021

- Developed and enhanced a cloud-native e-commerce platform using Java, Spring Boot, and React.js, reducing page load times by 35%, enabling smoother customer browsing and checkout experience.
- Integrated AI-based recommendation engine APIs within the platform to deliver personalized product suggestions, resulting in an 18% increase in cross-sell and upsell revenue and improving overall customer engagement metrics.
- Implemented real-time inventory management using AWS S3, DynamoDB, and MongoDB, enabling immediate stock updates, reducing stock-out errors by 25%, and improving supply chain reliability.
- Automated functional and regression testing using Selenium WebDriver and TestNG, increasing test coverage from 65% to 90% and reducing manual testing effort by 40%.
- Built and deployed microservices architecture on Kubernetes clusters with Jenkins and Docker pipelines, cutting deployment time by 40% and accelerating feature releases.
- Optimized MongoDB queries and indexing strategies, enhancing database performance and reducing average data retrieval time by 30%, ensuring rapid response for high-demand product pages under peak load conditions
- Coordinated with cross-functional teams in Agile sprints using Jira and Confluence, delivering 12+ sprint releases on schedule with zero critical production defects, ensuring stable releases.
- Built monitoring and logging dashboards using AWS CloudWatch and ELK Stack, achieving 99.9% uptime and reducing incident response time by 50%, supporting proactive system health management.

EDUCATION

Master of Science in Computer Science

University of Dayton, Ohio, USA

Bachelor of Technology in Computer Science and Engineering

Vidya Jyothi Institute of Technology, Telangana, India

CERTIFICATIONS

- Completed an Advanced Level Ethical Hacking Workshop at National Institute of Technology Warangal